1. Algebraic Conditioning and Support Selection

Some text here introducing the section and referencing Figure 1.

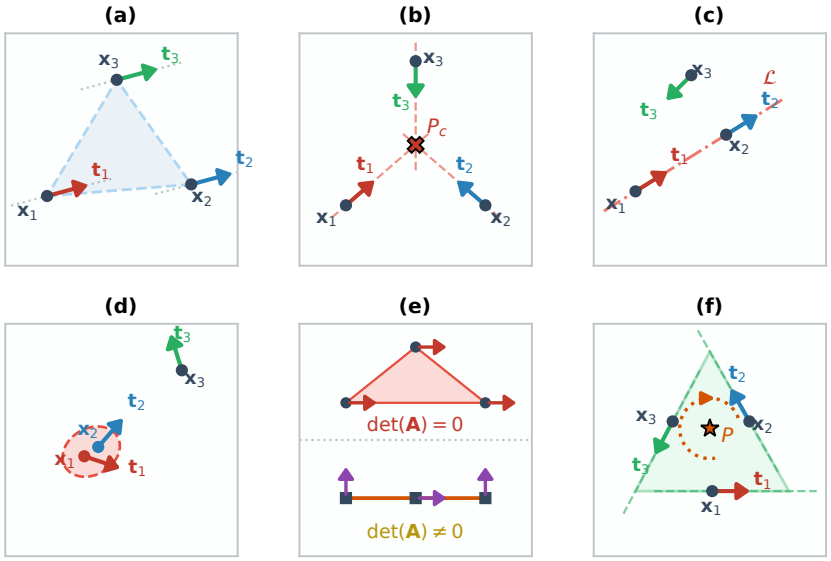

Figure 1: Geometric and directional singularity configurations of the VNMM 2D local placement matrix $\mathbf{A} \in \mathbb{R}^{3\times3}$ for the $\mathcal{L}^1$ solenoidal vector basis: (a) Global vector parallelism; (b) Concurrent lines of action; (c) Shared line of action; (d) Nodal coincidence; (e) Decoupling of spatial geometry and vector alignment; (f) Admissible and stable support.

1.1. Singularities in L1

To establish a systematic classification of algebraic singularities, we revisit the matrix:

$$\mathbf{A} = \begin{bmatrix} t_{1x} & t_{1y} & x_1 t_{1y} - y_1 t_{1x} & t_{2x} & t_{2y} & x_2 t_{2y} - y_2 t_{2x} & t_{3x} & t_{3y} & x_3 t_{3y} - y_3 t_{3x} \end{bmatrix} \tag{1}$$

More text here to see if it fits on the same page.